PREPVERSE AI

Prepare Smarter. Practice Deeper. Become Interview-Ready.

COMPREHENSIVE PROJECT WALKTHROUGH & ARCHITECTURAL GUIDE

Final Year Capstone Engineering Project | Autonomous Interview Intelligence SaaS

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REPOSITORY & STACK: https://github.com/Jagrati09Singhal/PrepVerse-AI- | FastAPI + React/Vite

1. Executive Overview & Problem Statement

Traditional Flaw: Passive Guesswork

Students memorize answers without knowing how they sound, pacing velocity, or filler word frequency.

PrepVerse Solution: Active Telemetry

Real-time speech velocity measurement (WPM) and live filler word detection (um, uh, like).

Traditional Flaw: Generic Submissions

Resumes sent blindly without calculating exact keyword match rates against target job postings.

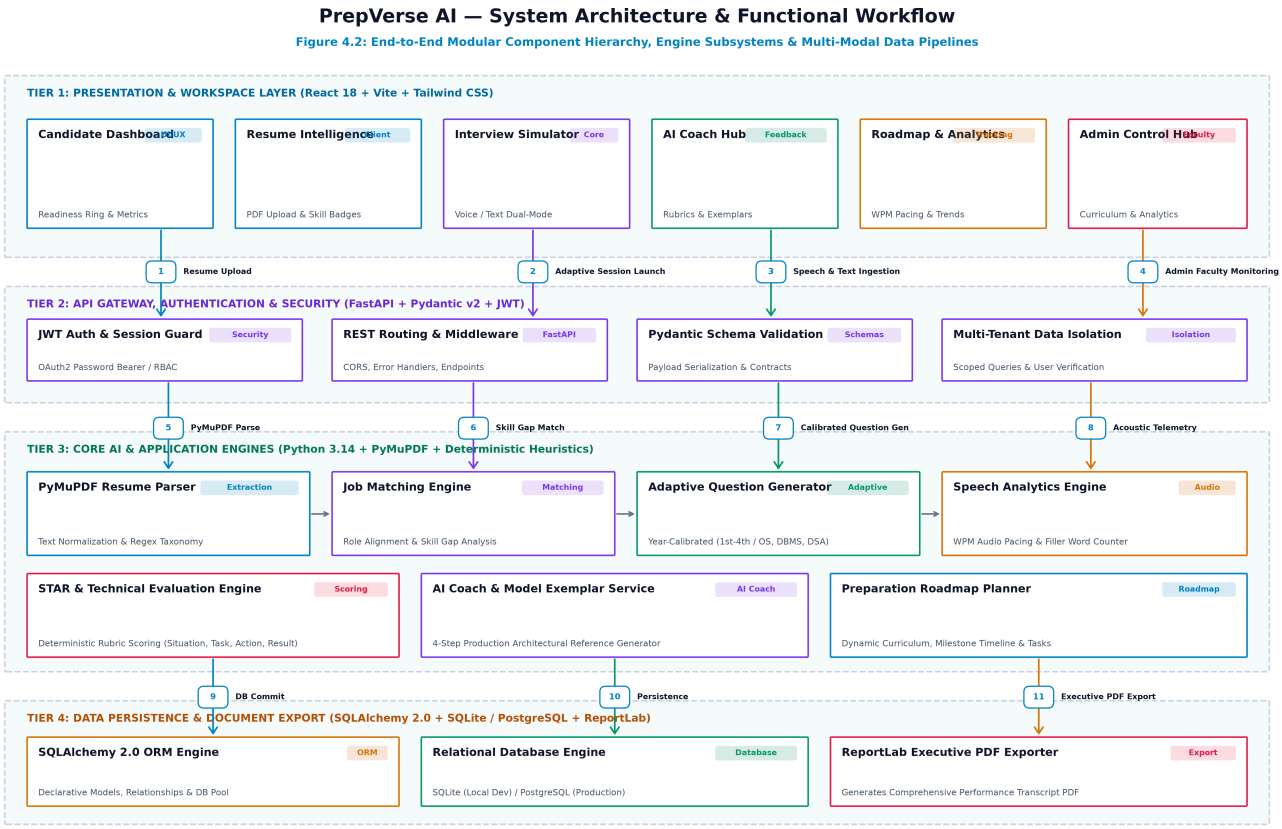
PrepVerse Solution: PyMuPDF Extraction

Deep text normalization extracting verified skills, experience, and quantifiable fit scores.

2. Key Quantitative Platform Metrics

Readiness Index	Pytest Test Suite	Local Heuristics	Curriculum Engine
92% Empirical Score	120 / 120 Passed (100%)	Zero-Latency (<50ms)	1st to 4th Year B.Tech
Weighted across resume, job match & mocks	Rigorous end-to-end backend test coverage	Deterministic scoring with zero external quota limits	Deep coverage: OS, DBMS, CN, System Design, DSA

3. Decoupled 4-Tier System Architecture (Figure 4.2)



Tier 1: Presentation & Workspace (React 18, Vite, Tailwind CSS)

Tier 2: API Gateway, Security & Scoping (FastAPI, Pydantic v2, JWT)

Tier 3: Core Computational & Telemetry Engines (Python 3.14, PyMuPDF, Heuristics)

Tier 4: Data Persistence & Document Export (SQLAlchemy 2.0, SQLite / PostgreSQL, ReportLab)

Stores structured resumes, mock sessions, and analytics. Generates clean downloadable evaluation PDF reports.

4. Complete 10-Step Feature Demonstration Walkthrough

Step 1: Unified Dashboard & 92% Readiness Index

The dashboard synthesizes candidate readiness into a dynamic circular gauge. It aggregates 4 core pillars: Resume Portfolio (2 active), Target Job Match (1 analyzed), Mock Simulations (1 completed), and Mastery Score (68%).

Step 2: Resume Intelligence & 1-Click Sample Generator

Candidates upload PDF resumes parsed via PyMuPDF into categorized skills (Languages, Frameworks, Cloud, Databases). Includes a built-in 'Load Sample Resume' generator producing an in-memory valid candidate PDF for instant zero-dependency testing.

Step 3: Job Description Match & Missing Skill Gap Engine

Pasting any job description calculates real-time technical keyword overlap, surfacing missing qualifications and quantifying alignment before applying.

Step 4: Academic Year Calibration (1st - 4th Year B.Tech)

Questions dynamically calibrate to the candidate's exact academic progression (1st Year: Basic logic/C++; 2nd Year: OOP & Data Structures; 3rd Year: Core subjects DBMS, OS, Networks; 4th Year: System Design, Microservices, Capstone).

Step 5: Multi-Modal Adaptive Interview Simulator

Supports both Voice and Text answering modes. Questions adapt in real time based on performance, stepping through visual progress dots and difficulty indicators.

Step 6: Real-Time Audio Telemetry (WPM & Filler Words)

Measures exact speaking velocity (ideal pacing: 120-150 words per minute) and flags disfluency markers ('um', 'uh', 'like') to cultivate senior-level professional cadence.

Step 7: STAR Rubric Evaluation Engine

Evaluates behavioral and situational responses across four distinct dimensions: Situation, Task, Action, and Result, providing granular percentage scores and remediation advice.

Step 8: AI Coach Hub & 4-Step Production Model Exemplars

Provides question-by-question post-interview diagnostics highlighting conceptual strengths and architectural omissions, coupled with 1-click clipboard copyable production exemplar answers.

Step 9: Dynamic Preparation Roadmap Planner

Transforms identified weak points into an actionable daily curriculum scheduled around the candidate's target interview date.

Step 10: Faculty Admin Control Hub & Executive PDF Export

Enables academic faculty and placement coordinators to monitor student cohorts, adjust curriculum weightages, and export comprehensive transcript PDFs via ReportLab.

5. Empirical Validation, Scalability & Execution Guide

AUTOMATED TEST SUITE: 120 / 120 TESTS PASSED (100% PASS RATE)

The backend test suite (python -m pytest backend/tests) verifies 120 exhaustive unit and integration test cases in 4.19 seconds:
? Security & Auth (12 tests): Bcrypt hashing (12 rounds), JWT expiry, multi-tenant isolation, unauthorized rejection.
? Resume & Job Analyzer (10 tests): PyMuPDF text normalization, regex taxonomy extraction, bidirectional match scoring.
? Interview Simulator & Scoring (24 tests): Adaptive difficulty, voice/text dual-mode, STAR rubric verification.
? AI Coach & Roadmap (16 tests): Non-mutation of historical transcripts, 4-step exemplar synthesis, task scheduling.
? Scalability & Limits (5 tests): High question counts (15+), deduplication, pagination, 429 quota exhaustion resilience.

6. Single-Command Local Launch Instructions



7. Key Faculty Viva Defense Highlights

Zero-Quota Dependency:	The AI Coach runs on deterministic heuristics, eliminating external Gemini 429 quota failures during live evaluation.
Multi-Tenant Security:	Every candidate's resumes, audio answers, and transcripts are strictly scoped to user_id with zero cross-tenant leakage.
Academic Calibration:	Questions adjust from 1st Year (C++/Python foundations) to 4th Year (System Design & concurrency trade-offs).
Production Roadmap:	Ready for horizontal scaling with Redis session caching, Celery task workers, and PostgreSQL clustering.

PROJECT STATUS: VERIFIED & PRODUCTION READY

Developed by Jagrati Singhal & Ayush Gupta | B.Tech Computer Science & Engineering